

Priyanshu Pramanik

Software Engineer

Kolkata, India | +91 6204058150 | priyanshualex@gmail.com
[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Analytical and detail-oriented Software Engineer specializing in full-stack web development, relational database design (PostgreSQL, MySQL), and intelligent automation. Proven capability in constructing high-performance React user interfaces, scalable Node.js/Express API backends, and Python-driven machine learning microservices. Strong academic record (8.95 CGPA) combined with technical internship experience designing data pipelines and automating workflows.

CORE TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python, SQL (PostgreSQL, MySQL), HTML5, CSS3
Frontend:	React, React 19, Tailwind CSS, DOM Manipulation, Responsive Web Design
Backend & APIs:	Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, MVC Architecture
Databases:	PostgreSQL, Neon DB, MySQL (indexing, schema design, query optimization), MongoDB
Tools & Cloud:	Git, GitHub, Docker, Postman, Vercel, Render, CI/CD pipelines
Systems & Auto:	ETL Pipelines, Data Validation (Pandas, NumPy), SMTP Outreach Automation

PROFESSIONAL EXPERIENCE

GAOTek Inc. — *Technical Intern*

Aug 2025 – Nov 2025

- Engineered an automated product documentation workflow to catalog 120+ SaaS/Automation products, reducing manual entry errors by over 20%.
- Integrated Apollo, Clay, and Thunderbird APIs to design a unified lead generation pipeline, generating 100+ qualified leads daily.
- Built and scaled email outreach campaigns using SMTP automation, maintaining database accuracy for follow-ups.

PROJECTS

Jarvis — *Local-First AI Assistant with Voice*

Python, FastAPI, Ollama, Voice AI

Tech Stack: Python, FastAPI, Ollama, faster-whisper, Piper TTS, openWakeWord, SQLite

- Architected a modular AI assistant OS where 7 isolated modules (brain, tools, memory, speech, wake, web, kernel) communicate exclusively through typed async events on an in-process event bus — zero inter-module imports — enabling independent replacement of any component.
- Engineered hands-free voice interaction combining wake-phrase detection ("hey jarvis") and clap detection (crest-factor signal analysis distinguishing impulsive claps from sustained speech) with local STT/TTS roundtrips, achieving full voice I/O with zero cloud APIs and zero cost.
- Built a tool-calling agent loop with per-conversation concurrency locks preventing token interleaving, gated conversation resets requiring explicit user confirmation before wiping history, and Ollama-native streaming with think:false for ~3x faster responses on CPU.
- Validated with a 30+ test suite using mock LLM providers, HTTP transports, and fake audio to ensure zero external dependencies in CI.

Wanna Watch — *AI-Powered Movie Discovery Platform*

React, Node, FastAPI, ML

Tech Stack: React 19, Node.js, FastAPI, Python, scikit-learn, PostgreSQL, Vercel, Render

- Designed a hybrid recommendation engine combining Content-Based Filtering (TF-IDF & Cosine Similarity) on a custom dataset with real-time TMDb API lookups.
- Formulated a 5-question personality quiz algorithm to output dynamic user matching scores, increasing user engagement.
- Developed full-stack user services featuring secure JWT authentication, stateful favorites/watchlists, and responsive lazy-loaded sections.
- Orchestrated a microservices-based deployment utilizing Vercel for frontend and Render for backend and machine learning microservices.

CleanFlow — *Automated Data Pipeline*

Python, FastAPI, ETL

Tech Stack: Python, FastAPI, Pandas, NumPy, RESTful APIs, Excel Reporting

- Engineered a robust ETL data cleaning pipeline to ingest, validate, and standardize messy real-world datasets, saving up to 85% of manual cleaning time.
- Implemented business-rule validation checks for format accuracy (emails, phone numbers, dates) and schema consistency.
- Programmed a reporting system that generates structured Excel summary reports mapping data anomalies for stakeholder review.

TaskForce — *Full-Stack Productivity Platform*

Node, Express, PostgreSQL, MySQL

Tech Stack: Node.js, Express, PostgreSQL, MySQL, Vanilla JS, REST API, JWT

- Overhauled a legacy task application into a high-performance, dark-themed brutalist UI featuring modular layout blocks and custom CSS typography.
- Scaled an MVC-based backend utilizing Express, optimizing database queries and connection pools across PostgreSQL and MySQL database instances.
- Implemented token-based authentication (JWT) and secure input validation to enforce server-side security.

EDUCATION

Swami Vivekananda University — *B.Tech in Computer Science & Engineering*

2024 - 2027 (Expected)

- Grade: CGPA: **8.95 / 10.0**

Government Polytechnic Khutri — *Diploma in Computer Engineering*

2021 - 2024

- Grade: **Distinction**

CERTIFICATIONS

CS50's Introduction to AI with Python

HarvardX (2025)

CS50's Introduction to Computer Science

HarvardX (2024)

Software Engineering

NPTel / IIT Kharagpur (2024)

OSINT Using Python

CyberVidyaPeeth (2023)